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**EEE401 – CAPSTONE PROJECT I**
**PROJECT PROPOSAL AND TWO-SEMESTER PROJECT PLAN**

# MIDIKLAVYE: A DIGITAL SUBTRACTIVE SOFTWARE SYNTHESIZER WITH A CLASSICAL-DSP PITCH-TO-MIDI INTERFACE

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**ABSTRACT**

MidiKlavye is a software project that couples a mathematically documented digital music synthesizer with a monophonic pitch-to-MIDI interface, using only classical digital signal processing (DSP). The synthesizer is built on one primary method, subtractive synthesis: band-limited oscillators generated with polynomial band-limited step (PolyBLEP) correction, a resonant second-order low-pass filter and a sample-accurate attack-decay-sustain-release (ADSR) envelope, all driven by MIDI note events. The pitch-to-MIDI interface estimates the fundamental frequency of whistled or hummed input with the YIN difference-function algorithm, compared against a normalised autocorrelation baseline, and converts the estimates to MIDI note events through silence gating, hysteresis, median smoothing and a minimum note duration. No neural networks or machine-learning models are used. This proposal reviews the supplied and independently checked literature, compares candidate synthesis and pitch-estimation methods, and defines the mathematical model, software architecture, requirements, verification experiments and risks. It also sets out a two-semester plan aligned with the Capstone Project 1 and 2 deadlines. Quantitative targets are defined as baseline experiments because no measurements exist yet. Frequency-modulation and wavetable synthesis, hardware-keyboard control and a graphical interface are scheduled as staged optional work after the subtractive core has been verified, whereas microtonal scales and polyphonic transcription are future work.

**Keywords:** *Digital signal processing, subtractive synthesis, pitch estimation, YIN algorithm, MIDI.*

# 1. Problem Statement

Digital synthesizers and audio-to-MIDI converters are everyday tools in electronic music, but for an engineering student they are rarely transparent. The oscillator, filter, envelope and pitch-tracking algorithms are usually consumed as black-box components. Understanding why a sawtooth oscillator aliases, why a resonant filter can become unstable, or why a pitch tracker reports a note an octave too low requires an implementation whose every equation, parameter and limitation can be inspected and measured.

Two coupled problems define this project. The first is **synthesis**: generating musically useful tones in discrete time without audible aliasing, with controllable timbre and a sample-accurate amplitude envelope, from standard MIDI note events. The second is **control**: converting a monophonic, non-stationary acoustic signal, such as a whistled or hummed melody, into the same note events. This requires estimating the fundamental frequency frame by frame, deciding when a note starts and ends, and rejecting unreliable estimates caused by silence, breath noise and octave errors.

Both problems must be solved under real-time constraints. A frame-based analyser needs a finite window of signal before an estimate exists, and audio blocks must be delivered on time to avoid dropouts. Rottondi et al. (2016) review perceptual latency-tolerance thresholds for

synchronous musical performance in the networked-performance setting; this motivates *measuring* the latency of the proposed chain rather than assuming it.

The problem addressed is therefore stated as follows: **can a modular system built only on classical DSP, with one primary synthesis method and a YIN-based monophonic pitch-to-MIDI interface, be implemented with all core algorithms written explicitly in Python, and be verified quantitatively (frequency accuracy, aliasing, envelope timing, pitch error, octave-error rate, note-segmentation errors and latency) within the two-semester capstone schedule?**

### 1.1 Practical significance

The project relates to the academic study of the problem in two ways. It applies established DSP theory (sampling, filtering, spectral analysis, difference-function pitch estimation) in a single integrated system, and it measures how that theory behaves on real whistling and humming recordings. Its practical value is a transparent reference implementation, with documented equations and repeatable experiments, that can be used for teaching and for later extension. The contribution is framed as engineering integration and verification. The proposal does not claim novelty over existing systems or superior performance to them.

## 2. Background

This section summarises the technical background and the literature reviewed so far. Sources were checked against publisher, repository or bibliographic-database records; Scopus and Web of Science indexing was *not* verified (see Section 2.7 and Appendix E). Machine-learning papers are discussed only to delimit the field and do not determine the implementation methodology.

### 2.1 Terminology and system boundary

A synthesizer renders a discrete-time signal x[n] from control events. In MIDI, a note is represented by a Note On and a Note Off event carrying a note number (0–127) and a velocity. Separating the event source from the sound engine lets the same synthesizer be driven by a keyboard, a file or a pitch tracker. Tanaka et al. (2023) use a MIDI interface in the same boundary role between a sensing front end and a synthesis back end, although their gesture mapping is learning-based and is not adopted here.

### 2.2 Digital sound synthesis and aliasing

Chowning (1973) introduced frequency modulation (FM) synthesis, in which the phase of a carrier is modulated by a second oscillator to produce a rich spectrum from few parameters. Rogozinsky et al. (2022) present detailed mathematical models of classical synthesis methods, and De Poli (2023) classifies sound models by their structure. Zhang et al. (2026) review string-instrument synthesis for interactive systems with a four-dimensional evaluation framework (fidelity, responsiveness, controllability and adaptability); they note that abstract digital methods appear less often in that literature because of their limited ability to reproduce fine instrument nuances. This is acceptable here, because MidiKlavye does not aim to imitate acoustic instruments; it aims at a controllable, explainable synthesis chain.

Subtractive synthesis generates a harmonically rich waveform and shapes it with a filter. Trivially sampled sawtooth and square waves contain discontinuities and therefore alias. Välimäki and Huovilainen (2007) review antialiasing oscillators for subtractive synthesis, and Välimäki et al. (2010) propose differentiated polynomial waveforms as an alias-suppressed alternative. Kleimola et al. (2010) show that phaseshaped oscillator waveforms are generally discontinuous and that aliasing can be reduced effectively by modifying samples around each discontinuity with the polynomial band-limited step (PolyBLEP) method. PolyBLEP is therefore a low-cost, explainable choice for the oscillator stage of this project.

### 2.3 Classical fundamental-frequency estimation

De Cheveigné and Kawahara (2002) describe YIN, an autocorrelation-based estimator with several modifications intended to prevent errors. They report error rates about three times lower than the best competing methods on a speech database recorded with a laryngograph, no upper limit on the frequency search range, and a method that is simple, low-latency and has few parameters. Mauch and Dixon (2014) extend the idea probabilistically (pYIN), which also provides a voiced/unvoiced decision. These results come from speech and general music material; they do not by themselves validate YIN on whistling or humming, which is why the project includes its own recordings and a baseline comparison (Section 4.6 and Appendix C).

### 2.4 Audio-to-MIDI conversion, evaluation and limits of transcription

Benetos et al. (2013) analyse the limitations of automatic music transcription and conclude that general-purpose methods do not capture the diversity of music signals; they suggest tailoring algorithms to specific use cases. The monophonic whistling and humming scenario of this project is such a restricted use case. Li et al. (2023) address the much harder polyphonic

problem with a constant-Q transform and shift-invariant probabilistic latent component analysis; this lies outside the scope of the project and is cited only to mark the boundary. For evaluation, Raffel et al. (2014) provide transparent implementations of common music-information-retrieval metrics, which informs the pitch and note metrics used in Appendix C.

### 2.5 Real-time operation and latency

Rottondi et al. (2016) review psycho-perceptual studies on latency tolerance for synchronous real-time performance and survey enabling technologies for networked music performance. Their context (remote ensembles) differs from a single-user synthesizer, so their thresholds are used only as discussion points, not as acceptance criteria. A simple calculation shows why latency must be measured: a 2048-sample YIN window lasts 46.4 ms at 44.1 kHz, and YIN also needs the longest searched lag in addition to the window. For humming with a lower limit of 80 Hz this lag is 551 samples (12.5 ms), so one estimate needs 2599 samples (58.9 ms) of signal and cannot appear earlier than that, independently of the audio block size. For whistling (lower limit 400 Hz) the extra lag is only 110 samples (2.5 ms), and a shorter window can be tried. Any sub-20 ms claim for the microphone path would therefore require a smaller window and a demonstrated accuracy cost; this is left to experiment T-R2. Brum and Moreno (2019) survey real-time singing-voice synthesis and are cited only as background on real-time constraints.

### 2.6 Learning-based work (context only)

Recent papers address score-to-audio synthesis with transformers (Dong et al., 2022), differentiable-DSP synthesizer sound matching (Masuda & Saito, 2023) and factorised neural codecs with ADSR control (Liu et al., 2025). They are relevant to the broader field but rely on trained models, and they are excluded from the implementation. The only element reused conceptually is the idea of comparing signals with spectral distances, which will be computed analytically in the analysis tools without any training.

### 2.7 Comparison of the reviewed literature and review status

Table 1 compares the 18 works cited in this proposal. Twelve are journal articles; none has yet been verified against Scopus or Web of Science. The guidelines require at least 25 papers from journals indexed in Scopus and/or Web of Science in the Chapter 2 literature survey (due in Fall week 10). This proposal therefore does **not** claim that this requirement is already met. A

targeted search plan with inclusion criteria is part of the timeline (Appendix E), and the database check is a manual step for the team.

**Table 1. Comparison of the literature cited in this proposal.**

| Reference | Type | Topic | Relevance | Role in this project |
|---|---|---|---|---|
| Benetos et al. (2013) | Journal | Limits of automatic music transcription | Context | Justifies the monophonic, use-case-specific scope |
| Brum & Moreno (2019) | Conference | Real-time singing voice synthesis survey | Low | Background on real-time synthesis constraints |
| Chowning (1973) | Journal | FM synthesis | Medium | Basis of the optional FM extension, Eq. (22) |
| de Cheveigné & Kawahara (2002) | Journal | YIN estimator | High | Primary pitch estimator, Eqs. (13)–(15) |
| De Poli (2023) | Journal | Structural taxonomy of sound models | Medium | Classification used in Section 4.2 |
| Dong et al. (2022) | Conference | Neural score-to-audio synthesis | Context only | Excluded (learning-based) |
| Kleimola et al. (2010) | Conference | Phaseshaping oscillators, PolyBLEP | High | Oscillator anti-aliasing, Eqs. (3)–(4) |
| Li et al. (2023) | Journal | Multi-pitch estimation (CQT, SI-PLCA) | Context only | Marks the polyphonic scope boundary |
| Liu et al. (2025) | Preprint | Neural synthesizer-style transfer with ADSR control | Context only | Excluded (learning-based) |
| Masuda & Saito (2023) | Journal | Differentiable synthesizer sound matching | Context only | Spectral-distance idea only |
| Mauch & Dixon (2014) | Conference | pYIN | Medium | Optional offline comparison |
| Raffel et al. (2014) | Conference | mir_eval metrics | High | Metric definitions, Eq. (20) |
| Rogozinsky et al. (2022) | Journal | Mathematical models of synthesis | High | Notation for subtractive and FM models |
| Rottondi et al. (2016) | Journal | Latency tolerance, networked performance | Medium | Latency discussion (different context) |

| Reference | Type | Topic | Relevance | Role in this project |
|---|---|---|---|---|
| Tanaka et al. (2023) | Journal | EMG-driven instrument with MIDI interface | Low | Precedent for MIDI as system boundary |
| Välimäki & Huovilainen (2007) | Journal | Antialiasing oscillators | High | Aliasing background |
| Välimäki et al. (2010) | Journal | Differentiated polynomial waveforms | Medium | Possible alternative to PolyBLEP |
| Zhang et al. (2026) | Journal | Review of string synthesis for interactive systems | Medium | Evaluation dimensions; synthesis taxonomy |

Within the sources reviewed so far, none reports a measured, end-to-end evaluation of a classical monophonic pitch-to-MIDI chain on whistling and humming driving an explicit software synthesizer. This observation is limited to the current pool and will be re-examined in the Week 10 survey; it is not presented as a research gap or novelty claim. The supplied literature inventory contained several bibliographic inconsistencies, listed in Appendix E, which were corrected before citing.

## 3. Project Goals and Objectives

### 3.1 Aim

The aim is to design, implement and verify a modular Python software system that (i) synthesizes tones with one explicit, mathematically documented subtractive synthesis chain driven by MIDI events and (ii) converts monophonic whistling or humming into MIDI note events with classical DSP, and to report measured accuracy, stability and latency.

### 3.2 Objectives

- **O1 – Synthesis engine.** Implement band-limited PolyBLEP oscillators and a resonant biquad low-pass filter, and document the discrete-time equations, parameters and expected output.
- **O2 – Envelope control.** Implement a sample-accurate ADSR envelope and define its behaviour for Note On and Note Off events, including early release.

- **O3 – Classical pitch estimation.** Implement the YIN estimator with a normalised-autocorrelation baseline for monophonic input; pYIN may be used as an optional offline comparison.
- **O4 – Pitch-to-MIDI conversion.** Map valid frequency estimates to MIDI notes (A4 = 440 Hz) and generate Note On/Off events through silence gating, hysteresis, smoothing and minimum duration.
- **O5 – Analysis and verification.** Provide FFT/STFT analysis and repeatable experiments for synthesis correctness, pitch estimation, segmentation and real-time behaviour (Appendix C).
- **O6 – Integration.** Integrate the core modules into a working prototype that runs offline on WAV files and, if schedule permits, in real time from a microphone.
- **O7 – Literature and documentation.** Deliver the proposal, the literature survey with at least 25 verified Scopus/WoS journal papers, the review of alternative methods, the interim report and the final report on the official deadlines.

### 3.3 Research questions

- **RQ1.** By how much does PolyBLEP correction reduce measurable aliasing compared with a naively sampled waveform across the playable note range, and at what computational cost?
- **RQ2.** How accurately and stably does YIN estimate the fundamental frequency of whistled and hummed input compared with a normalised autocorrelation baseline (cent error, octave-error rate, noise robustness)? The working hypothesis, to be tested and not assumed, is that YIN produces fewer octave errors than the baseline.
- **RQ3.** Which segmentation parameters give an acceptable trade-off between missed notes and false note events?
- **RQ4.** What end-to-end latency does the real-time chain exhibit on the team's hardware as a function of audio block size and analysis frame length? If the real-time mode is not reached, RQ4 is answered with the algorithmic latency computed from Eq. (23) and the measured processing times.

### 3.4 Scope classification

Table 2 classifies the features proposed in the technical design document. No feature has been declared mandatory by the supervisor; the classification below is an engineering recommendation based on the time available and the grading emphasis on clear, non-imaginary objectives. Optional items are scheduled in the weekly plan (Section 6) but are attempted only

after the core they depend on has been verified. The architecture allows additional oscillator types to be added without rewriting the core.

**Table 2. Scope classification of proposed features.**

| Feature | Category | Rationale |
|---|---|---|
| Subtractive synthesis (PolyBLEP oscillator, biquad filter, ADSR) | Core | Primary synthesis method (Section 4.2) |
| MIDI event model, router, QWERTY virtual keyboard | Core | Hardware-free input; needed by every test |
| 8-voice polyphony with simple voice stealing | Core | Small additional code; supports keyboard play |
| YIN and ACF pitch estimation, pitch-to-MIDI, segmentation | Core | Second technical axis of the project |
| Offline WAV processing, WAV and MIDI file export | Core | Repeatable experiments and deliverables |
| FFT / STFT analysis, spectrograms, metrics | Core | Verification (O5) |
| Real-time microphone input | Should (stretch) | Offline processing is the fallback if latency or dropouts cannot be controlled |
| USB MIDI keyboard input (velocity, pitch bend, CC, sustain) | Optional (scheduled) | Fall TW9; aftertouch only if time permits |
| FM oscillator with Bessel sideband check | Optional (scheduled) | Second synthesis method; Fall TW8, after the primary chain is verified at the TW7 checkpoint |
| GUI (on-screen piano, parameter panel, live $f_0$ curve) and MIDI Learn | Optional (scheduled) | Fall TW10 and Spring SW7; useful for demonstration, not needed for verification |
| Quantisation modes beyond nearest semitone (scale-lock, continuous pitch bend) | Optional (scheduled) | Spring SW5; nearest-semitone mapping is core |
| Filter envelope, LFO, extra modulation, 16 voices | Optional | Only after the core is verified |
| Wavetable synthesis (table, interpolation, morphing, mipmap) | Optional (stretch) | Fall TW8 (stretch), offline only; dropped if the checkpoint is not passed or FM overruns |
| Microtonal scales (including the undefined "Safir" scale) | Future work | Requires a formal definition first |

| Feature | Category | Rationale |
|---|---|---|
| Audio effects, advanced preset management | Future work | Outside the verification focus |
| Polyphonic transcription, speech-to-music mode | Future work | Outside the classical monophonic scope (Section 2.4) |

### 3.5 Exclusions

The project excludes any neural-network, deep-learning or machine-learning component, including pretrained models such as CREPE; reliable polyphonic or chord transcription; transcription of arbitrary speech into notes; and the design of physical MIDI or audio hardware.

## 4. Methods

### 4.1 Research framework

In terms of the options listed in the guidelines, the study combines **simulation modelling** (discrete-time models of the oscillator, filter, envelope and pitch estimator implemented in software), **conceptual modelling** (the architecture and the mathematical specification) and **quantitative analysis** (repeatable experiments with defined metrics). Every experiment is planned, not completed; no results are reported in this document. The work proceeds in four phases: literature review and requirements (Fall), method comparison and feasibility prototypes (Fall), implementation and integration (Spring), and final verification and reporting (Spring).

### 4.2 Comparison of candidate methods and selection

#### *4.2.1 Primary synthesis method*

Table 3 compares the three candidate methods against the criteria set for this project. The ratings are qualitative engineering judgements, not measurements, and they are open to supervisor review.

**Table 3. Qualitative comparison of candidate primary synthesis methods.**

| Criterion | FM synthesis | Subtractive synthesis | Wavetable synthesis |
|---|---|---|---|
| Mathematical depth | Medium: phase modulation, Bessel-function sidebands | High: oscillator aliasing theory plus filter theory (z-transform, poles, stability) | Medium: table design, interpolation error |

| Criterion | FM synthesis | Subtractive synthesis | Wavetable synthesis |
|---|---|---|---|
| Implementation complexity | Low | Medium: PolyBLEP, biquad design, parameter-dependent stability | Medium to high: table generation, band-limited table sets |
| Ease of explaining equations | High | High (oscillator and filter equations are short) | Medium |
| Audio quality and controllability | Rich spectra; parameters interact non-intuitively | Intuitive controls (cut-off, resonance, envelope) | Depends on stored tables |
| Aliasing behaviour | Sidebands may exceed the Nyquist frequency at high index | Discontinuities alias; PolyBLEP reduces it (Kleimola et al., 2010) | Aliases when pitch-shifted without band-limited tables |
| Verification options | Sideband amplitudes versus Bessel values | Filter magnitude response; alias-to-signal ratio; envelope timing | Interpolation error; harmonic content |
| Literature in the supplied pool | Chowning (1973); Rogozinsky et al. (2022) | Kleimola et al. (2010); Välimäki & Huovilainen (2007); Välimäki et al. (2010) | Little in the pool |
| Relevance to an EEE degree | Medium | High: filtering and stability are core EEE topics | Medium |
| Time and risk | Low risk | Moderate risk, fits two semesters | Higher risk |

**Recommendation: subtractive synthesis** (PolyBLEP-corrected sawtooth and square oscillators, a resonant second-order low-pass filter and an ADSR amplitude envelope) is proposed as the single primary method. It offers the broadest mathematical content for an Electrical and Electronics Engineering project, with both oscillator aliasing and filter stability to analyse, and it is well supported by the literature in hand. FM synthesis and wavetable synthesis are scheduled as staged optional extensions (Fall TW8): FM can be added as another oscillator and checked analytically against Bessel values, and a wavetable oscillator can be checked for interpolation error. Both are attempted only after the subtractive chain has been verified at the TW7 checkpoint. This recommendation is provisional until confirmed by the supervisor.

### *4.2.2 Primary pitch-estimation algorithm*

Table 4 compares the classical candidates for monophonic whistling and humming. Whistling is close to a pure tone with a very strong fundamental, while humming has a harmonic spectrum, and the two modes need different search ranges.

**Table 4. Classical fundamental-frequency estimators considered.**

| Method | Principle | Typical failure cases | Cost | Role |
|---|---|---|---|---|
| Autocorrelation (ACF) | Peak of normalised autocorrelation, Eq. (16) | Octave errors; picks sub-multiples | $O(W \cdot \tau_{max})$ | Baseline |
| AMDF | Magnitude difference, no multiplications | Similar to ACF; sensitive to level changes | $O(W \cdot \tau_{max})$ | Not implemented |
| YIN | Squared difference, cumulative mean normalisation, absolute threshold, parabolic interpolation (de Cheveigné & Kawahara, 2002) | Noise; rapid pitch change within a frame; threshold choice | $O(W \cdot \tau_{max})$ | Primary |
| pYIN | Probabilistic threshold distributions with smoothing (Mauch & Dixon, 2014) | Needs more parameters; offline smoothing | Higher | Optional offline comparison |
| FFT peak / harmonic analysis | Spectral peak or harmonic comb | Resolution $\Delta f = f_s/N$; harmonic confusion | $O(N \log N)$ | Cross-check in analysis tools |

**Recommendation:** YIN as the primary estimator, the normalised autocorrelation as the baseline that justifies YIN's added steps, and an FFT-peak cross-check in the analysis tools. AMDF is not implemented because it adds little insight beyond the ACF/YIN pair. pYIN is an optional offline comparison; the team considers it acceptable under the classical-only restriction (a probabilistic but untrained method), subject to confirmation by the supervisor.

## 4.3 Mathematical model

This section defines the equations to be implemented. Symbols are defined at first use; all frequencies are in Hz, times in seconds and levels in dB relative to full scale (dBFS) unless stated. Standard DSP results without a journal citation (for example the biquad design relations) will be referenced to textbooks in the literature survey.

### *4.3.1 Discrete-time fundamentals*

A continuous-time signal $x_c(t)$ sampled at rate $f_s$ gives x[n], with sample index n, and no frequency above the Nyquist frequency $f_s/2$ can be represented. An oscillator of frequency $f_0$ is generated by a phase accumulator φ[n] in [0, 1) with increment Δ per sample:

$$x[n] = x_c(n/f_s), \quad f_{Nyq} = f_s/2 \tag{1}$$

$$\varphi[n] = (\varphi[n-1] + \Delta) \bmod 1, \quad \Delta = f_0/f_s \tag{2}$$

### *4.3.2 Band-limited oscillator and resonant filter*

The naive sawtooth 2φ[n] − 1 has a discontinuity at each phase wrap and aliases. The PolyBLEP-corrected sawtooth subtracts a two-sample polynomial residual r around the wrap (Kleimola et al., 2010). A square wave uses the same residual at both of its edges, and a triangle can be obtained by integration if required:

$$s_{saw}[n] = 2\varphi[n] - 1 - r(\varphi[n], \Delta) \tag{3}$$

$$r(t, \Delta) = 2t/\Delta - (t/\Delta)^2 - 1 \text{ for } 0 \le t < \Delta; \quad (t-1)^2/\Delta^2 + 2(t-1)/\Delta + 1 \text{ for } 1-\Delta < t < 1; \quad 0 \text{ otherwise} \tag{4}$$

The oscillator output is shaped by a second-order (biquad) low-pass filter with cut-off frequency $f_c$ and quality factor Q. Its transfer function and the corresponding difference equation are:

$$H(z) = (b_0 + b_1 z^{-1} + b_2 z^{-2}) / (1 + a_1 z^{-1} + a_2 z^{-2}) \tag{5}$$

$$y[n] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2] - a_1 y[n-1] - a_2 y[n-2] \tag{6}$$

The coefficients follow from the bilinear-transform design with $\omega_0 = 2\pi f_c/f_s$ and $\alpha = \sin \omega_0/(2Q)$:

$$\omega_0 = 2\pi f_c/f_s, \quad \alpha = \sin \omega_0 / (2Q) \tag{7}$$

$$b_0 = b_2 = (1 - \cos \omega_0)/(2a_0), \quad b_1 = (1 - \cos \omega_0)/a_0, \quad a_0 = 1 + \alpha, \quad a_1 = -2\cos \omega_0/a_0, \quad a_2 = (1 - \alpha)/a_0 \tag{8}$$

The filter is stable when both poles of H(z) lie inside the unit circle. This is verified numerically over the permitted range of $f_c$ and Q (requirement MR-03), because time-varying parameters can still cause audible artefacts even when each fixed setting is stable. For this design ($0 < \omega_0 < \pi$, $Q > 0$) every fixed setting is stable, so the cut-off is clamped below the Nyquist frequency (Table 5) and the test concentrates on time-varying parameters. Aliasing will be quantified as the energy of spectral components that are not harmonics of $f_0$ relative to the harmonic energy.

### *4.3.3 ADSR envelope*

The envelope e[n] in [0, 1] is a four-state machine driven by Note On and Note Off. With attack time $T_a$, decay time $T_d$, sustain level S in [0, 1] and release time $T_r$, the piecewise-linear updates per sample are:

$$\textit{Attack:}\quad e[n] = \min(1,\ e[n-1] + 1/(T_a f_s)) \tag{9}$$

$$\textit{Decay:}\quad e[n] = \max(S,\ e[n-1] - (1 - S)/(T_d f_s)) \tag{10}$$

$$\textit{Release:}\quad e[n] = \max(0,\ e[n-1] - e_r/(T_r f_s)) \tag{11}$$

Here $e_r$ is the envelope value at the instant of Note Off, so releasing during attack or decay starts from the current level without a jump. Zero-length stages are handled by clamping the step to one sample. The output of voice k with velocity $v_k$ (scaled to [0, 1]) and the mixed output with master gain g are:

$$y_k[n] = v_k \cdot e_k[n] \cdot (h_k * s_k)[n],\quad y[n] = g \sum_k y_k[n] \tag{12}$$

Here $s_k$ is the oscillator signal, $h_k$ the filter impulse response and * denotes convolution (implemented recursively by Eq. 6). Samples with $|y[n]| > 1$ are counted as clipping events, which are reported in the synthesis tests.

### *4.3.4 YIN pitch estimation and autocorrelation baseline*

For a frame $x_j$ of W samples and lag $\tau$ in samples, YIN (de Cheveigné & Kawahara, 2002) computes the difference function, its cumulative mean normalised version, and then selects the first lag whose normalised value falls below a threshold $\theta$ and is a local minimum. Parabolic interpolation around the selected lag gives a sub-sample period estimate, and $1 - d'(\hat{\tau})$ can serve as a periodicity confidence:

$$d_t(\tau) = \sum_{j=1}^{W} (x_j - x_{j+\tau})^2 \tag{13}$$

$$d'_t(\tau) = 1 \text{ for } \tau = 0;\quad d_t(\tau) / [\ (1/\tau) \sum_{j=1}^{\tau} d_t(j)\ ] \text{ otherwise} \tag{14}$$

$$\hat{\tau} = \tau + [d'(\tau-1) - d'(\tau+1)] / [2(d'(\tau-1) - 2d'(\tau) + d'(\tau+1))],\quad \hat{f_0} = f_s/\hat{\tau} \tag{15}$$

The baseline is the normalised autocorrelation, whose largest peak in the allowed lag range gives the period:

$$\rho_t(\tau) = \sum x_j x_{j+\tau} / \sqrt{(\sum x_j^2 \cdot \sum x_{j+\tau}^2)} \tag{16}$$

The integration window must span at least two periods of the lowest frequency of interest, so $W \geq 2f_s/f_{min}$, and each estimate needs $W + \tau_{max}$ samples of signal, where $\tau_{max} = f_s/f_{min}$ is the

longest lag searched. The search ranges for humming and whistling are provisional (Table 5) and will be set from the recordings.

#### *4.3.5 Pitch-to-MIDI mapping and note segmentation*

With the tuning reference A4 = 440 Hz mapped to MIDI note 69, a valid estimate $f > 0$ maps to the continuous note number m, the nearest integer note n and the deviation $\delta$ in cents. The inverse gives the nominal frequency of a note:

$$m = 69 + 12\log_2(f/440), \quad n = \mathrm{round}(m), \quad \delta = 100(m - n) \tag{17}$$

$$f_n = 440 \cdot 2^{(n-69)/12} \tag{18}$$

Zero, negative, non-finite or low-confidence estimates are rejected before mapping. The frame level $L_t$ (rms over N samples) drives the silence gate:

$$L_t = 20\log_{10}\left(\sqrt{(1/N)\,\Sigma_n\, x[n]^2}\right) \quad [\mathrm{dBFS}] \tag{19}$$

Segmentation then proceeds as follows. A note starts when $L_t$ exceeds the on-threshold $L_{on}$ and the estimate is valid for M consecutive frames. It ends when $L_t$ falls below the lower off-threshold $L_{off}$ (hysteresis) or the estimates are invalid for $M_{off}$ frames. The note number is median-filtered over K frames, and a note change is accepted only if the new number persists for $M_{ch}$ frames and exceeds the active note by a margin, which prevents flicker between neighbouring semitones. Notes shorter than $T_{min}$ are discarded, and a repeated note requires a level dip between onsets. Velocity is fixed by default; mapping it to level is an optional refinement.

#### *4.3.6 Spectral analysis and error metrics*

Spectra are computed with a windowed discrete Fourier transform over frames of N samples and hop H. The frequency resolution $\Delta f = f_s/N$ is a limit on the peak position, and a spectral peak is not necessarily the fundamental, which is why FFT peaks serve only as a cross-check. Pitch error is expressed in cents relative to a reference $f_{ref}$:

$$e = 1200\log_2(\hat{f}_0 / f_{ref}) \quad [\mathrm{cents}] \tag{20}$$

$$X[m, k] = \Sigma_{n=0}^{N-1}\, w[n]\, x[n + mH]\, e^{-j2\pi kn/N}, \quad \Delta f = f_s/N \tag{21}$$

A frame counts as correct when $|e| \le 50$ cents and as an octave error when $|e - 1200q| \le 50$ cents for an integer $q \neq 0$, in line with the metric conventions of Raffel et al. (2014). The optional FM oscillator, and the audio latency budget, are:

$$y[n] = A\sin(2\pi f_c n/f_s + I\sin(2\pi f_m n/f_s)) \tag{22}$$

$$L_{tot} = L_{in} + L_{frame} + L_{proc} + L_{out}, \quad T_B = B/f_s \tag{23}$$

In Eq. (22), $f_c$ and $f_m$ are the carrier and modulator frequencies and I is the modulation index; sidebands appear at $f_c \pm kf_m$ with amplitudes proportional to the Bessel functions $J_k(I)$ (Chowning, 1973). In Eq. (23), B is the audio block size, $T_B$ the block duration (256 samples at 48 kHz give 5.33 ms), and $L_{frame}$ is up to $W + \tau_{max}$ (the integration window plus the longest searched lag) if an estimate is emitted when the frame is complete. Each term will be measured, not assumed.

### *4.3.7 Provisional parameters*

Table 5 lists the parameters and how each will be fixed. All values are starting points and are not performance claims.

**Table 5. Provisional parameters and how they will be set.**

| Parameter | Symbol | Provisional value | How it will be fixed |
|---|---|---|---|
| Sampling rate | $f_s$ | 44.1 or 48 kHz | Audio-device test in WP1; both supported |
| Audio block size | B | 256 samples | Latency and underrun experiments at several sizes (T-R1, T-R2) |
| YIN frame length | W | 2048 samples (humming); 1024 to be tested for whistling | Sensitivity analysis; $W \geq 2f_s/f_{min}$, and the buffer per estimate is $W + \tau_{max}$ |
| Hop size | H | 256–512 samples | Latency/accuracy trade-off (T-P5) |
| Search range, humming | $f_{min}$–$f_{max}$ | 80–1000 Hz | Set from recordings |
| Search range, whistling | $f_{min}$–$f_{max}$ | 400–3000 Hz | Set from recordings |
| YIN threshold | $\theta$ | 0.1 | Sweep 0.05–0.3 |
| Gate thresholds | $L_{on}$, $L_{off}$ | Noise floor + 10 dB; 6 dB lower | Noise floor measured per recording session |
| Median length | K | 5 frames | Sweep 3–9 |
| Minimum note duration | $T_{min}$ | 60 ms | Sweep 40–120 ms |
| Tuning reference | A4 | 440 Hz (MIDI 69) | Fixed by convention |
| Voices | $N_v$ | 8 | Fixed for the core; 16 is optional |
| Filter cut-off | $f_c$ | 20 Hz to 0.45 $f_s$ (clamped below Nyquist) | Clamped in code; swept, also with fast modulation, in test T-S3 |
| Filter quality factor | Q | 0.5–10 | Stability test MR-03 |

### 4.4 Software architecture

Figure 1 shows the architecture. The sound engine accepts only normalised MIDI events, so the keyboard, the pitch tracker and an optional USB device are interchangeable sources. The real-time path (audio callback) performs only block synthesis from preallocated buffers, with no file access, logging or blocking locks. Pitch analysis, file output, plotting and the optional GUI run outside the callback and communicate through bounded queues with an explicit overflow policy that discards the oldest analysis frame. Offline processing (WAV to MIDI, event list to WAV) reuses the same module functions without audio hardware, which makes it deterministic and testable.

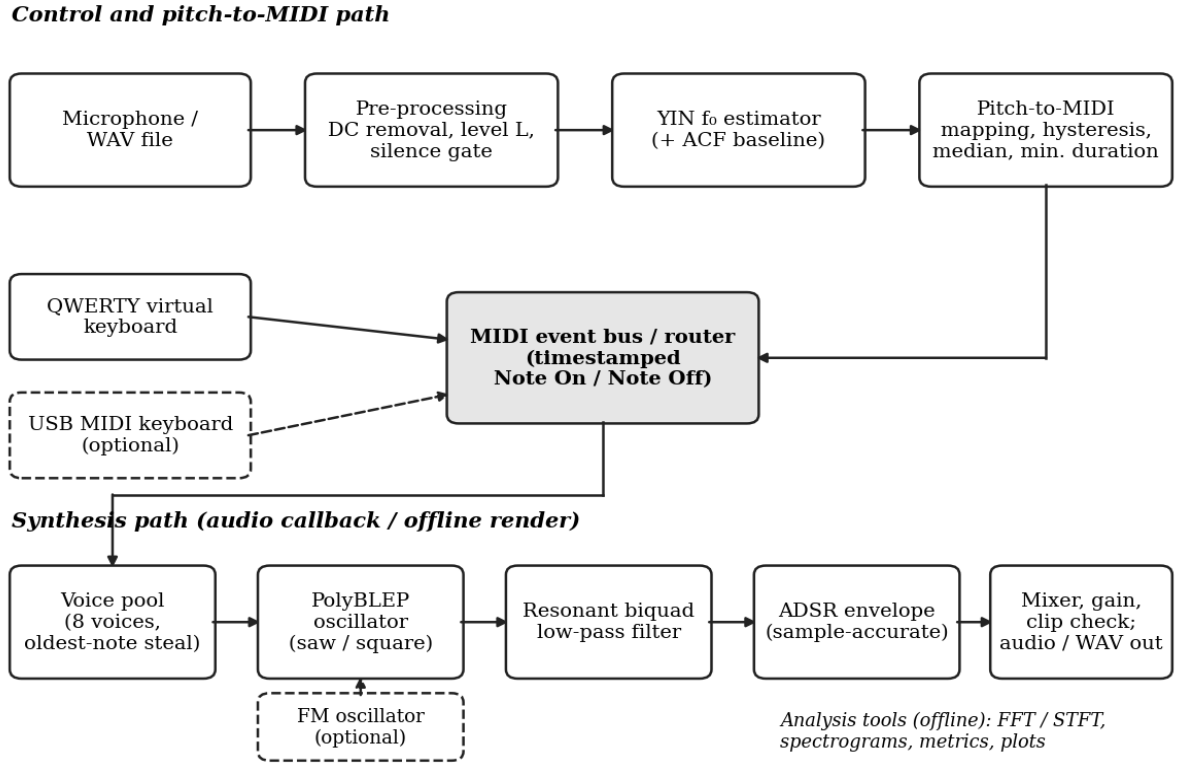



**Fig. 1. Software architecture and data flow of MidiKlavye (dashed boxes are optional).**

Table 6 lists the modules, their responsibilities and interfaces. Each module is a pure, GUI-independent function or class so that it can be tested in isolation (Appendix C).

**Table 6. Software modules, responsibilities and interfaces.**

| Module | Responsibility | Interface and data |
|---|---|---|
| core/events | MIDI event types, validation, timestamps, router | Event(type, note, velocity, channel, t) |
| synth/osc | Phase accumulator, PolyBLEP saw and square (FM optional) | render(freq, n_samples) returns float array |
| synth/filters | Biquad coefficient design and filtering | process(block) with persistent state |

| Module | Responsibility | Interface and data |
|---|---|---|
| synth/envelope | ADSR state machine | gate_on(), gate_off(), next_block(n) |
| synth/engine | Voice pool, voice stealing, mixing, clipping count | handle(event), render_block(B) |
| pitch/preprocess | DC removal, framing, level $L_t$, silence gate | frames and level arrays |
| pitch/yin, pitch/acf | Frame-wise $f_0$ and confidence | estimate(frame) returns (f0, conf) |
| pitch/segmenter | Mapping, hysteresis, median, minimum duration | frames to list of note events |
| inputs | QWERTY keyboard, microphone capture, WAV reader (USB MIDI optional) | event queue or frame queue |
| analysis | FFT/STFT, spectrogram, metrics, plots, export | functions on arrays; report files |
| tests | Unit, integration and regression tests | pytest suites and experiment scripts |

### 4.5 Implementation environment

**Python** is the primary environment, as preferred by the team, and it has adequate audio, MIDI and plotting libraries. MATLAB is available and may be used for independent cross-checks of selected results, such as the filter magnitude response, but is not part of the product. The core algorithms (oscillators, PolyBLEP, biquad, ADSR, YIN, ACF, segmentation, frequency-to-MIDI mapping) are written explicitly and are not taken from libraries; libraries are used for numerical arrays, device I/O, plotting and test support (Table 7). Library implementations such as *scipy.signal* or *librosa* are used only as independent references when validating the custom code.

**Table 7. Proposed dependencies and why each is needed.**

| Dependency | Purpose | Custom or library |
|---|---|---|
| NumPy | Array computation, FFT | Library; algorithms on top are custom |
| SciPy | Reference filter responses and spectrograms for validation | Library, validation only |
| sounddevice (PortAudio) | Audio output and microphone capture | Library (device I/O) |
| mido, python-rtmidi | MIDI file export; optional USB MIDI input | Library (device and file I/O) |
| soundfile or wave | WAV reading and writing | Library |

| Dependency | Purpose | Custom or library |
|---|---|---|
| Matplotlib | Spectra, spectrograms, result plots | Library |
| pytest | Automated tests | Library |
| mir_eval (optional) | Cross-check of metric implementations | Library, validation only |
| PySide6 (optional) | Minimal GUI if the optional extension is approved | Library |

### 4.6 Data collection and organisation

The project needs test signals with known ground truth and recordings of real whistling and humming (Table 8). No company data, questionnaire or personal data other than the team's own voice recordings is planned, so the permissions required for company or questionnaire data in the guidelines are not expected to apply; this will be confirmed with the supervisor, and recordings of other people would be made only with the supervisor's guidance and any required approval.

**Table 8. Planned data sets (quantities are provisional).**

| Set | Content | Provisional size | Ground truth and purpose |
|---|---|---|---|
| D1 | Synthetic sinusoids and harmonic tones, with and without white noise (SNR from clean to 0 dB) | About 100 frequencies, 6 noise levels | Known $f_0$; accuracy and noise robustness (T-P1 to T-P3) |
| D2 | Synthetic note sequences with controlled onsets, vibrato and glides | About 50 sequences | Known onsets and offsets; segmentation tests (T-N1, T-N2) |
| D3 | Guided recordings of humming and whistling by the team, in a quiet and a noisier room | At least 40 recordings, about 10 minutes | Reference note sequence played by the synthesizer plus manual onset and offset annotation from spectrograms (T-P4) |

For D3 the ground truth is the reference melody the performer follows, aligned by hand, so it carries annotation uncertainty that will be reported. Recording conditions (microphone, distance, room, sampling rate) are documented with each file so that experiments are repeatable.

### 4.7 Verification, sensitivity analysis and requirements

Appendix A lists the requirements with identifiers, priorities, verification methods and acceptance criteria; Appendix B the work packages; Appendix C the tests; and Appendix D the

risk register. Where no baseline exists, acceptance criteria are marked provisional and the final threshold is set after a baseline experiment in Spring weeks 1–2, using the values achieved by the baseline method and the application need (a note should not be mis-assigned to a neighbouring semitone). Following the guidelines' recommendation, results will include comparative analyses (YIN versus ACF, PolyBLEP versus naive oscillator) and sensitivity analyses over noise level, frame length, YIN threshold, hysteresis margin and median length.

# 5. Expected Outcomes

## 5.1 Deliverables

- A Python application and source code implementing the core modules in Table 6, with offline WAV processing, WAV and MIDI export and, if the schedule permits, real-time operation from a microphone.
- Automated tests, experiment scripts, the recorded data sets D1–D3 with annotations, and a README that allows the experiments to be repeated.
- The written deliverables on the official deadlines: this proposal, the Introduction, the literature survey (at least 25 verified journal papers), the review of alternative methods, the CP1 interim report, and in Spring the final methodology, results, discussion, conclusion and final report with presentation.

## 5.2 Engineering contributions

The contribution is an integrated, documented and measured classical-DSP implementation: explicit equations with symbols and assumptions, a verified synthesis chain, a quantified comparison of YIN and autocorrelation on whistling and humming, and a measured latency profile. It is not claimed to be novel or to outperform existing systems.

## 5.3 Planned experimental results

The following quantities will be measured and reported with tables and plots: oscillator frequency accuracy; alias-to-signal ratio for PolyBLEP versus naive oscillators; filter response and stability; envelope timing; median absolute pitch error in cents, valid-estimate rate and octave-error rate for YIN and ACF; robustness to noise; onset error, offset error, false notes and missed notes; processing time, underruns and mic-to-sound latency at several block sizes. No values are stated here because none have been measured.

### 5.4 Known limitations

Classical pitch estimators assume a single, approximately periodic source. Octave errors, noise sensitivity and fast pitch changes within a frame are known failure modes, and speech with voiced and unvoiced segments, breath noise or several simultaneous sources will not be transcribed reliably (Benetos et al., 2013). PolyBLEP leaves residual aliasing near the Nyquist frequency (Kleimola et al., 2010). Frame-based analysis imposes an algorithmic latency of the order of the frame length (Eq. 23), and Python real-time audio has to be validated empirically on the team's hardware.

### 5.5 Flexibility and extension

The architecture isolates the oscillator behind a simple interface, so FM and wavetable oscillators can be added without changing the voice pool or the pitch path, and the tuning map is a separate function so that another scale could be supported after a formal definition. If real-time performance is not achievable, the project falls back to offline processing without changing the core. If new assumptions arise from the recordings (for example a different search range), only the configuration parameters in Table 5 change.

### 5.6 Outside the current scope and future work

Microtonal scales (including the "Safir" scale once formally defined), audio effects, advanced preset management, polyphonic transcription and a speech mode are listed as future work in Table 2. Optional items that are not reached remain future work.

## 6. Timeline

The plan integrates the week-by-week roadmap supplied for this project (twelve technical weeks per semester) with the official deadlines of Capstone Projects 1 and 2. Technical weeks are labelled TW1–TW12 (Fall) and SW1–SW12 (Spring) and are mapped to the official week numbers in Tables 9 and 10. The Fall semester ends in January and the Spring semester ends on 30 April 2027; week numbers are to be mapped to the academic calendar. In the Fall, weeks 1–4 are used for the proposal, requirements and set-up, so TW1–TW12 run in official weeks 5–16 and the Term I report and demonstration fall in week 16, which is the interim-report deadline. In the Spring, SW1–SW12 coincide with official weeks 1–12, and weeks 13–14 are reserved for supervisor corrections and the jury presentation, ending on 30 April 2027. The online courses and quizzes (weeks 3–7) and the literature survey (weeks 5–10) run in parallel with TW1–TW6. A feasibility prototype of the pitch path (offline ACF and YIN on synthetic

tones) is built in TW7, so that the highest-risk part of the project is tested before the Spring semester.

Three adaptations were made to the roadmap and are flagged for confirmation (Appendix F). First, CREPE is read only as machine-learning context in the literature work and is **not** integrated, because the implementation is restricted to classical methods; the algorithm comparison in SW3 is YIN, autocorrelation and pYIN. Second, subtractive synthesis, the primary method, is scheduled before FM, polyphony (a core feature) is moved ahead of the optional work (to TW6), and FM, wavetable, the hardware keyboard, the GUI and MIDI Learn are marked optional, to be attempted only if the primary chain passes the TW7 checkpoint. Third, the official results deadline (Spring week 8) falls before the systematic experiments of SW9, so the week 8 submission reports the design, implementation and preliminary results, and the systematic experiments feed the discussion due in week 10.

**Table 9. Fall semester plan (Term I: synthesizer and MIDI control).**

| Week | Off. week | Topic | Output | Official deliverable |
|---|---|---|---|---|
| — | 1–4 | Proposal, requirements baseline, repository, audio-device test; online courses start (weeks 3–7) | Approved proposal | Proposal (week 4) |
| TW1 | 5 | Literature (Chowning 1973, MIDI 1.0, YIN; CREPE as context only), Python environment, git repository, keyboard input | Literature notes, repository | — |
| TW2 | 6 | DSP foundations: sampling, aliasing, phase-accumulator oscillator, audio output | Script that plays and records a sine | — |
| TW3 | 7 | ADSR envelope: state machine, linear and exponential curves | Envelope plots, voice class | Introduction (week 7) |
| TW4 | 8 | Architecture: event queue, router, audio callback; virtual keyboard v1 | Monophonic play from the QWERTY keyboard | — |
| TW5 | 9 | **Subtractive synthesis (primary):** PolyBLEP, resonant filter; filter envelope if time permits | Aliasing experiment (T-S2, T-S3) | — |
| TW6 | 10 | *Polyphony: voice pool, voice stealing, performance* | Real-time play with 8 voices (16 optional) | Literature survey (week 10) |

| Week | Off. week | Topic | Output | Official deliverable |
|---|---|---|---|---|
| TW7 | 11 | *Checkpoint: primary chain verified (T-S1 to T-S5). Analysis tools (FFT/STFT, alias-to-signal ratio). Feasibility prototype: offline ACF and YIN on synthetic tones* | Primary chain verified; first pitch-error plot on synthetic tones | — |
| TW8 | 12 | Optional: FM synthesis: C:M ratio, index envelope, Bessel spectrum; wavetable (stretch, offline) only if FM is finished early | FM sounds and spectrum comparison (T-S6); wavetable offline if the checkpoint is passed | — |
| TW9 | 13 | *Optional:* hardware keyboard: velocity, pitch bend, CC, sustain; aftertouch if time permits | Keyboard control | — |
| TW10 | 14 | *Optional:* GUI v1: on-screen piano, parameter panel, MIDI Learn | Usable interface | Review of alternative models and methods (week 14) |
| TW11 | 15 | Analysis tools, synthesis comparison experiments, latency measurement (buffer week) | Comparison table | — |
| TW12 | 16 | Term I report and demonstration | Report and live demonstration | Interim report (week 16, January) |

**Term I delivery:** a software synthesizer with a verified subtractive chain, ADSR envelope and polyphony that can be played from the virtual keyboard (and from a hardware keyboard and with FM and wavetable oscillators if the optional steps are reached); the report contains the literature, the architecture and the synthesis comparison.

**Table 10. Spring semester plan (Term II: from microphone to melody, and evaluation).**

| Week | Off. week | Topic | Output | Official deliverable |
|---|---|---|---|---|
| SW1 | 1 | Feedback; pitch-detection literature; labelled test data (synthetic, own recordings, and an open data set only if its licence and ground truth are verified) | Test set (D1–D3) | — |
| SW2 | 2 | Write YIN from scratch, validate on synthetic signals, compare with pYIN; baseline experiment for thresholds | Validated YIN | — |

| Week | Off. week | Topic | Output | Official deliverable |
|---|---|---|---|---|
| SW3 | 3 | YIN, autocorrelation and pYIN comparison (no CREPE) | Algorithm comparison table (T-P1, T-P2) | — |
| SW4 | 4 | Pitch-to-MIDI: conversion, median filter, hysteresis, velocity | Offline .wav to .mid | Final methodology (week 4) |
| SW5 | 5 | Quantisation: nearest semitone (core); scale-lock and continuous pitch bend (optional); audio to MIDI to synthesis chain | Offline end-to-end system | — |
| SW6 | 6 | Real-time microphone mode, latency budget. **Checkpoint** | Live microphone to synthesizer | — |
| SW7 | 7 | *Optional:* mixed mode, detected note shown on the on-screen piano, live $f_0$ curve | Integrated GUI | — |
| SW8 | 8 | Input/output comparison tools: $f_0$, spectrogram, mir_eval metrics; MFCC if time permits | Comparison module | Model implementation and results (week 8) |
| SW9 | 9 | Systematic experiments: source × algorithm × quantisation; latency; sensitivity analysis | Result tables | — |
| SW10 | 10 | Hardening, platform tests, unit tests, user manual (buffer week); code freeze | Stable release | Discussion (week 10) |
| SW11 | 11 | Final report writing | Report draft | — |
| SW12 | 12 | Final presentation rehearsal, demonstration, delivery of the report; weeks 13–14 for corrections and the jury presentation | Final delivery | Conclusion (week 12); final report and presentation (week 14, 30 April 2027) |

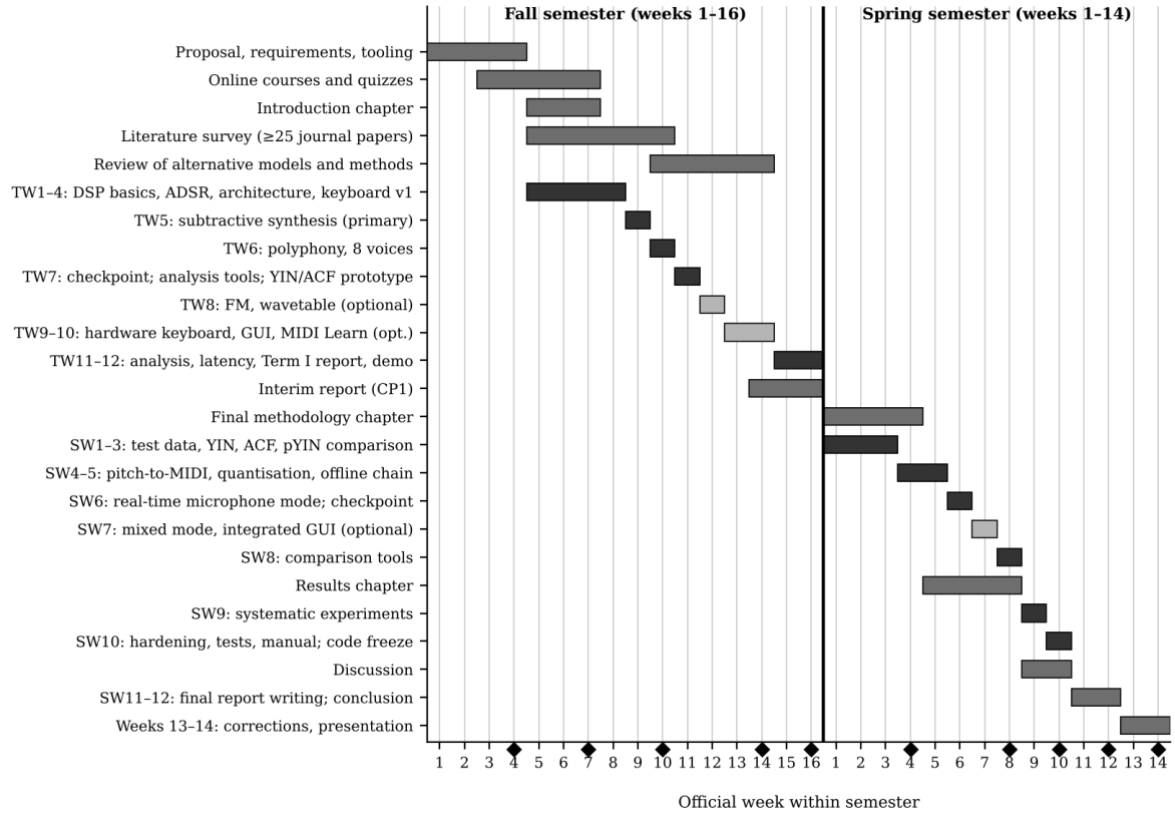



**Fig. 2. Gantt chart over the two semesters (light grey bars are optional or parallel activities; darker bars are core work; diamonds mark the official deadlines).**

Both team members contribute to every work package; leads per package will be agreed and recorded in the interim report. The main schedule risks are the Fall workload (courses, a 25-paper journal survey and implementation in parallel), real-time audio stability and the compressed time between the end of the experiments and the results deadline. Their mitigations and contingencies are in Appendix D.

## Appendix A. Requirements Specification

Priority uses Must / Should / Could. A criterion marked (P) is provisional: no baseline measurement exists yet, so the final threshold is set after the baseline experiment in Spring weeks 1–2 as described in Section 4.7. Requirements are grouped as functional (FR), performance (PR), mathematical (MR), usability (UR) and documentation (DR).

**Table A1. Requirements specification.**

| ID | Description | Category | Priority | Rationale | Verification | Acceptance criterion | Depends on |
|---|---|---|---|---|---|---|---|
| FR-01 | MIDI event model with Note On/Off, note (0–127), velocity (1–127), channel, timestamp | Functional | Must | Common interface for all inputs | Unit test | All 128 notes round-trip; invalid values rejected | — |
| FR-02 | Band-limited sawtooth and square oscillators (PolyBLEP) | Functional | Must | Primary synthesis method (O1) | T-S1, T-S2 | Fundamental within ±1 cent of target for MIDI notes 36–96 (P) | FR-01 |
| FR-03 | Resonant biquad low-pass filter with cut-off and Q | Functional | Must | Timbre shaping (O1) | T-S3 | Magnitude response matches theory; stable for all permitted parameters | FR-02 |
| FR-04 | Sample-accurate ADSR with early release | Functional | Must | Envelope control (O2) | T-S4 | Stage durations within one sample of specification; no jump at early release | FR-01 |
| FR-05 | 8-voice polyphony with oldest-note stealing | Functional | Must | Keyboard play | Unit test, random sequences | No stuck notes in 1000 random sequences; ninth note steals the oldest | FR-01, FR-04 |
| FR-06 | QWERTY virtual keyboard generating MIDI events | Functional | Must | Hardware-free input | Demonstration | Key down/up produce matching Note On/Off | FR-01 |

| ID | Description | Category | Priority | Rationale | Verification | Acceptance criterion | Depends on |
|---|---|---|---|---|---|---|---|
| FR-07 | Deterministic offline rendering of an event list to WAV | Functional | Must | Repeatable experiments | T-S5 | Identical output for identical input on the same machine | FR-02 to FR-04 |
| FR-08 | MIDI file export of generated note events | Functional | Must | Inspect pitch-to-MIDI output | Reload test | Exported file reloads with identical notes and times (within tick resolution) | FR-01 |
| FR-09 | YIN estimator returning $f_0$ and confidence per frame | Functional | Must | Primary estimator (O3) | T-P1, T-P2 | Implements Eqs. (13)–(15); agrees with a reference library on test frames within a tolerance set in WP9 | — |
| FR-10 | Normalised autocorrelation baseline estimator | Functional | Must | Needed for comparison (RQ2) | T-P2 | Implements Eq. (16); same interface as FR-09 | — |
| FR-11 | Frequency-to-MIDI mapping with A4 = 440 Hz; invalid estimates rejected | Functional | Must | Pitch-to-MIDI (O4) | Unit test | Eqs. (17)–(18) exact for all notes; zero, negative and NaN inputs rejected | FR-09 |
| FR-12 | Note segmentation with gate, hysteresis, median, minimum duration | Functional | Must | Stable note events (O4) | T-N1, T-N2 | Parameters configurable; behaviour defined for repeated notes | FR-11 |
| FR-13 | Offline WAV-to-MIDI pipeline | Functional | Must | Core of the evaluation | T-P4, T-N1 | Processes all D2 and D3 files without manual steps | FR-09, FR-12 |
| FR-14 | Real-time microphone to synthesizer | Functional | Should | Live demonstration (O6) | T-R2, T-I1 | Runs for 10 minutes without a crash; falls back to offline mode if not achieved | FR-12, FR-05 |
| FR-15 | FFT/STFT, spectrogram and metric tools | Functional | Must | Verification (O5) | Unit test, review | Results agree with a reference library within numerical tolerance | — |

| ID | Description | Category | Priority | Rationale | Verification | Acceptance criterion | Depends on |
|---|---|---|---|---|---|---|---|
| FR-16 | USB MIDI keyboard input with velocity, pitch bend, CC and sustain | Functional | Could | Optional extension (TW9) | Demonstration | Hardware events produce correct notes and controller changes | FR-01 |
| FR-17 | FM oscillator (carrier, modulator, index) | Functional | Could | Optional extension (TW8) | T-S6 | Sideband amplitudes within a tolerance of Bessel values (P) | FR-02 |
| FR-18 | GUI with on-screen piano, parameter panel and live $f_0$ curve | Functional | Could | Optional extension (TW10, SW7) | Demonstration | Works without affecting audio performance | FR-14 |
| FR-19 | Wavetable oscillator with interpolation (offline) | Functional | Could | Optional extension (TW8) | T-S7 | Interpolation error and aliasing reported (P) | FR-02 |
| FR-20 | MIDI Learn for controller-to-parameter assignment | Functional | Could | Optional extension (TW10) | Demonstration | A controller can be assigned to a parameter | FR-16 |
| PR-01 | No audio underruns in real-time synthesis with 8 voices at the chosen block size | Performance | Should | Audio quality | T-R1 | 0 underruns in 10 minutes (P); otherwise report the smallest stable block size | FR-05 |
| PR-02 | Measured mic-to-sound latency reported at three or more block sizes | Performance | Must | RQ4 | T-R2 | At least 20 trials per setting with mean and standard deviation; threshold set after baseline If FR-14 is not achieved, the algorithmic latency (the frame and block terms of Eq. (23)) and the measured processing time are reported instead. | FR-14; offline fallback FR-13 |
| PR-03 | Pitch accuracy on clean synthetic signals | Performance | Must | RQ2 | T-P1 | Median absolute error $\leq 10$ cents (P) | FR-09 |

| ID | Description | Category | Priority | Rationale | Verification | Acceptance criterion | Depends on |
|---|---|---|---|---|---|---|---|
| PR-04 | Octave-error rate reported for YIN and ACF on the same data | Performance | Must | RQ2 | T-P2, T-P4 | Both rates reported; YIN versus ACF comparison stated with the data | FR-09, FR-10 |
| PR-05 | Segmentation errors reported | Performance | Must | RQ3 | T-N1, T-N2 | Onset/offset error, false and missed notes reported; thresholds set after baseline | FR-12 |
| PR-06 | Aliasing reduction of PolyBLEP versus naive oscillator | Performance | Must | RQ1 | T-S2 | Alias-to-signal ratio reported per note; improvement quantified in dB | FR-02 |
| MR-01 | Every implemented equation documented with symbols, units and assumptions | Mathematical | Must | Explainability | Report review | Section 4.3 matches the code | — |
| MR-02 | Frequency-to-MIDI mapping verified for all notes | Mathematical | Must | Correctness | Unit test | Round trip within $10^{-9}$ relative error | FR-11 |
| MR-03 | Filter numerical stability | Mathematical | Must | Resonant filters can be unstable | T-S3 | Bounded output over $10^7$ random samples across the parameter range; poles inside the unit circle; repeated with a rapidly modulated cut-off | FR-03 |
| UR-01 | One command per offline demonstration | Usability | Should | Reproducibility | Review | README steps reproduce the demonstration | FR-07, FR-13 |
| UR-02 | Human-readable configuration file | Usability | Should | Parameter tuning | Review | $f_s$, B and Table 5 parameters set without code changes | — |

| ID | Description | Category | Priority | Rationale | Verification | Acceptance criterion | Depends on |
|---|---|---|---|---|---|---|---|
| DR-01 | Documented code, README and experiment scripts | Documentation | Must | Repeatability | Review | A new user reproduces one experiment from the README | — |
| DR-02 | Reports delivered on the official deadlines | Documentation | Must | Grading | Supervisor approval | Uploads on or before each deadline | — |
| DR-03 | Reference list in APA with at least 25 verified Scopus/WoS journal papers in the CP1 survey | Documentation | Must | Official requirement | Manual database check | Each paper checked and recorded in Appendix E | — |

## Appendix B. Work Breakdown Structure

Durations use the technical-week labels of Tables 9 and 10 (TW = Fall, SW = Spring) with official week numbers where useful. Packages marked (optional) are attempted only after the packages they depend on are verified. Both team members contribute to every package; leads will be assigned by the team.

**Table B1. Work packages.**

| ID | Objective and detailed tasks | Depends on | Duration | Deliverables | Acceptance and verification | Main risks |
|---|---|---|---|---|---|---|
| WP0 | Courses and administration. Online courses and quizzes; group and supervisor actions. | — | Official weeks 3–7 | Course and quiz records | Records uploaded | Time conflict with other courses |
| WP1 | Requirements and tooling. Requirements baseline; repository; Python environment; audio and MIDI device test; choose $f_s$ and initial block size; literature notes (TW1). | — | Weeks 1–5 (TW1) | Requirements (Appendix A), environment, device report | Supervisor approval of proposal (week 4) | Device or driver problems |

| ID | Objective and detailed tasks | Depends on | Duration | Deliverables | Acceptance and verification | Main risks |
|---|---|---|---|---|---|---|
| WP2 | Literature review. Search plan, database verification, at least 25 journal papers, comparison table, gap discussion; review of alternative models and methods. | WP1 | Weeks 5–14 | Introduction (week 7), survey (week 10), method review (week 14) | Each paper verified and recorded; supervisor approval | Too few indexed journal papers (R11) |
| WP3 | Synthesis foundations and architecture. DSP basics, phase-accumulator oscillator, audio output, ADSR, event queue, router, audio callback, virtual keyboard v1. | WP1 | TW2–TW4 | Voice class, event bus, QWERTY play | FR-01, FR-04, FR-06; T-S4, T-S5 | Callback glitches; integration (R7, R10) |
| WP4 | Subtractive synthesis and polyphony. PolyBLEP oscillators, resonant filter, filter envelope (optional), voice pool and voice stealing, performance tests. | WP3 | TW5–TW6 | Verified primary chain; 8-voice real-time play | FR-02, FR-03, FR-05; T-S1 to T-S3; T-R1 | Filter instability; aliasing (R6) |
| WP5 | Optional synthesis extensions. FM with Bessel check; wavetable with interpolation, morphing, mipmap (offline); checkpoint at TW7. | WP4 verified | TW8 | FM and wavetable oscillators | FR-17, FR-19; T-S6, T-S7 | Scope growth (R1) |
| WP6 | Optional input devices and interface. Hardware keyboard (velocity, pitch bend, CC, sustain), GUI v1, MIDI Learn; integrated GUI with live $f_0$ curve in SW7. | WP4 | TW9–TW10, SW7 | Keyboard control; usable GUI | FR-16, FR-18, FR-20 | Device compatibility (R8) |
| WP7 | Synthesis evaluation and Term I report. Analysis tools, synthesis comparison experiments, latency measurement, interim report and demonstration. | WP4, WP5 | TW11–TW12 | Comparison table; interim report (week 16) | T-S1 to T-S5 executed; supervisor approval | Time (R13) |
| WP8 | Test data. D1 and D2 generated; D3 recorded and annotated; open data set only if verified. | WP1 | SW1 | Data sets with metadata | Annotated files reviewed by both members | Noisy rooms; annotation effort |

| ID | Objective and detailed tasks | Depends on | Duration | Deliverables | Acceptance and verification | Main risks |
|---|---|---|---|---|---|---|
| WP9 | Pitch estimation. YIN from scratch, validation against a reference library, ACF baseline, pYIN comparison, thresholds from the baseline experiment. A first offline ACF and YIN prototype on synthetic tones is built earlier, in TW7. | WP8 | TW7 (prototype); SW2–SW3 | Validated pitch module; comparison table | FR-09, FR-10; T-P1, T-P2 | Octave errors; noise (R2, R3, R4) |
| WP10 | Pitch-to-MIDI. Mapping, median filter, hysteresis, velocity, minimum duration; MIDI export; quantisation modes; offline end-to-end chain. | WP9, WP4 | SW4–SW5 | Offline .wav to .mid and chain to synthesis; final methodology (week 4) | FR-11 to FR-13; T-N1, T-N2 | Segmentation errors (R5) |
| WP11 | Real-time microphone mode. Audio capture, thread and queue design, latency budget; mixed mode (optional). | WP10 | SW6–SW7 | Live microphone to synthesizer | FR-14; T-R1, T-R2, T-I1 | Dropouts, latency (R7) |
| WP12 | Evaluation and hardening. Comparison module ($f_0$, spectrogram, mir_eval, MFCC optional); systematic experiments; sensitivity analysis; unit and platform tests; manual. | WP9–WP11 | SW8–SW10 | Results (week 8), result tables, stable release | All Must tests executed and reported | Insufficient time (R13) |
| WP13 | Reporting and presentation. Final methodology, results, discussion, conclusion, final report, presentation and demonstration. | All | SW1–SW12; weeks 13–14 | Reports on deadlines; final presentation (30 April 2027) | Supervisor approvals; APA compliance | Late writing; format errors |

## Appendix C. Verification and Evaluation Plan

These are planned tests, not completed experiments, and no results are stated. Where acceptance criteria are provisional (P), the threshold is fixed after the baseline experiment (Section 4.7). Pitch metrics follow Section 4.3.6.

**Table C1. Planned tests.**

| ID | Objective | Input | Procedure | Metric | Acceptance criterion | Expected evidence |
|---|---|---|---|---|---|---|
| T-S1 | Generated frequency versus target | Notes 36–96 at fixed velocity, 2 s each | Render, window, locate spectral peak with interpolation | Frequency error (cents) | ≤ 1 cent (P) | Table and plot |
| T-S2 | Aliasing of PolyBLEP versus naive saw/square | Same notes, both oscillators | Compute energy of non-harmonic components relative to harmonic energy | Alias-to-signal ratio (dB) | Improvement quantified per note (RQ1) | Plot versus note; spectra |
| T-S3 | Filter response and stability | Impulse and random input; sweep of $f_c$ and Q | Compare measured magnitude response with theory; run $10^7$ samples; repeat with a fast cut-off sweep | Deviation (dB); boundedness | No unbounded output; deviation within tolerance (P) | Response plots |
| T-S4 | ADSR timing and early release | Gate sequences with extreme settings | Measure stage durations; release during attack and decay | Duration error (samples); step size | Within one sample; no jump | Envelope plots |
| T-S5 | Determinism and clipping | Fixed event list | Render twice; count $\|y\| > 1$ | Identical output; clip count | Identical; zero clipping at default gain (P) | Hash comparison |
| T-S6 | FM sidebands (optional) | Carrier, modulator, index sweep | Compare sideband amplitudes with $\|J_k(I)\|$ | Amplitude error | Within tolerance (P) | Bar plots |
| T-S7 | Wavetable interpolation (optional) | Table with 2048 samples, notes across the range | Compare linear interpolation with a high-resolution reference; measure aliasing | Interpolation error; alias-to-signal ratio | Reported (P) | Error plots |
| T-P1 | Pitch accuracy on clean tones | D1 without noise | Run YIN and ACF | Median absolute error (cents) | ≤ 10 cents (P) | Error table |
| T-P2 | Octave errors, YIN versus ACF | D1 harmonic tones, including strong second harmonic | Run both estimators on identical frames | Octave-error rate | Both rates reported and compared | Comparison table |

| ID | Objective | Input | Procedure | Metric | Acceptance criterion | Expected evidence |
|---|---|---|---|---|---|---|
| T-P3 | Noise robustness | D1 at SNR from clean to 0 dB | Run both estimators per SNR | Median error; valid rate; octave rate versus SNR | Curves reported; threshold set after baseline | Robustness curves |
| T-P4 | Real humming and whistling | D3 | Run pipeline; compare with annotated reference | Raw pitch accuracy; octave rate; valid-estimate rate | Reported with uncertainty; threshold set after baseline | Per-file and overall table |
| T-P5 | Stability across frames and sensitivity | D1 and D3 with parameter sweeps | Vary W, H, θ, K | Frame-to-frame jitter; accuracy | Trade-off curves reported | Sensitivity plots |
| T-N1 | Segmentation accuracy | D2 and D3 | Compare detected notes with reference | Onset and offset error (ms); false and missed notes | Thresholds set after baseline | Event tables |
| T-N2 | Repeated notes and minimum duration | D2 with repeated notes and very short notes | Run pipeline | Repeated triggers; dropped short notes | Behaviour matches specification | Annotated timeline |
| T-R1 | Real-time synthesis load | 8 voices, block sizes 128 to 1024 | Run 10 minutes per setting | Processing time per block; CPU use; underruns | 0 underruns at the chosen size (P) | Load table |
| T-R2 | Mic-to-sound latency | Pulse or tone input; loopback measurement | At least 20 trials per block size and frame length | Mean and standard deviation of delay (ms) | Reported; target set after baseline | Latency table |
| T-I1 | End-to-end integration | Live humming to synthesized output | Run for 10 minutes, record output | Crash-free run; qualitative note correctness | Completes without failure or documented fallback | Recording and log |

## Appendix D. Risk Register

Likelihood (L) and impact (I) are scored from 1 (very low) to 5 (very high) by the team; the score is L × I. The bands are provisional: 15 or more is high, 8 to 14 is medium, and 7 or less is low. Scores are judgements and not measurements, and they will be reviewed at each milestone.

**Table D1. Risk register.**

| ID | Risk | L | I | Score | Mitigation | Contingency | Monitoring stage |
|---|---|---|---|---|---|---|---|
| R1 | Excessive scope | 4 | 4 | 16 | Single primary synthesis method; optional items only after the primary chain is verified at the TW7 checkpoint | Drop optional items; keep core | Every milestone; TW7 and SW6 checkpoints |
| R2 | Poor pitch accuracy on real recordings | 3 | 4 | 12 | Baseline experiments; confidence gate; tune W and θ | Restrict to whistling or to clean recordings and document | SW2–SW5 |
| R3 | Octave errors | 4 | 3 | 12 | YIN normalisation; FFT-peak cross-check; search ranges per mode | Median filtering; report the error rate openly | T-P2, T-P4 |
| R4 | Noise sensitivity | 3 | 3 | 9 | Gate calibrated to measured noise floor; light band limiting | Restrict to quiet-room recordings | T-P3 |
| R5 | Segmentation errors (false or missed notes) | 4 | 3 | 12 | Hysteresis, minimum duration, sensitivity analysis | Relax parameters; report trade-off | T-N1, T-N2 |
| R6 | Aliasing at high notes | 2 | 3 | 6 | PolyBLEP; measure alias-to-signal ratio | Limit note range or add oversampling | T-S2 |
| R7 | Real-time dropouts or high latency | 3 | 4 | 12 | Preallocated buffers; no work in callback; test several block sizes | Fall back to offline pipeline | WP11; T-R1, T-R2 |
| R8 | MIDI device compatibility | 2 | 2 | 4 | QWERTY input as default; USB MIDI optional | Skip USB MIDI | WP6 (TW9) |

| ID | Risk | L | I | Score | Mitigation | Contingency | Monitoring stage |
|---|---|---|---|---|---|---|---|
| R9 | Dependency or packaging problems | 3 | 3 | 9 | Pinned versions; minimal libraries; environment file | Offline-only mode without audio library | WP1; each month |
| R10 | Integration delays | 3 | 4 | 12 | Interface definitions early; continuous integration of tests | Reduce voices; simplify GUI | TW6; SW5–SW7 |
| R11 | Fewer than 25 Scopus/WoS journal papers found | 4 | 4 | 16 | Search plan (Appendix E); start in F5; include related-area journals | Agree with supervisor how to treat borderline papers | F7, F10 |
| R12 | Unverified or wrongly cited references | 3 | 4 | 12 | Verify each paper against publisher and database; record in Appendix E | Remove the paper from the list | Every citation addition |
| R13 | Insufficient time for evaluation and documentation (results deadline in week 8 precedes SW9 experiments) | 3 | 5 | 15 | Code freeze in SW10; writing in parallel; preliminary results in the week 8 report; sensitivity runs scripted early | Reduce experiments to Must tests | SW5, SW8, SW10 |
| R14 | Fall overload (courses, 25-paper survey and implementation in parallel) | 4 | 3 | 12 | Optional steps (TW8, TW9, TW10) are skipped first; buffer week TW11 | Move optional work to Spring or drop it | TW3, TW7, TW11 |
| R15 | Uneven teamwork or coordination | 2 | 3 | 6 | Shared repository, code review, written package leads | Re-plan leads at milestones | Each milestone |

## Appendix E. Literature Verification Records and Search Plan

Only abstracts, repository records or bibliographic-database entries were read for the sources below; full texts have not been read, and no claim of full-text review is made. Scopus and Web of Science status was not checked in this draft and must be checked manually for every journal.

## E.1 Verification record of the cited sources

**Table E1. Verification record.**

| Reference | Venue and type | Identifier | Checked against | Scopus / WoS status |
|---|---|---|---|---|
| Benetos et al. (2013) | J. Intell. Inf. Syst.; journal | DOI 10.1007/s10844-013-0258-3 | Aggregator record with DOI and abstract; issue and pages recalled (†) | Not verified — manual database verification required. |
| Brum & Moreno (2019) | SBCM 2019; conference | DOI 10.5753/sbcm.2019.10422 | SBC OpenLib record with citation | Conference; not counted toward the journal quota |
| Chowning (1973) | J. Audio Eng. Soc.; journal | Not retrieved | Seen only in other papers' reference lists and the team's draft (†) | Not verified — manual database verification required. |
| de Cheveigné & Kawahara (2002) | J. Acoust. Soc. Am.; journal | DOI 10.1121/1.1458024 | DOI record and abstract | Not verified — manual database verification required. |
| De Poli (2023) | Musica/Tecnologia; journal | DOI 10.36253/music_tec-14417 | Journal website (abstract, pages 7–75) | Not verified — manual database verification required. |
| Dong et al. (2022) | ICASSP 2022; conference | arXiv:2202.06034 | arXiv abstract; pages from dblp (†) | Conference; not counted |
| Kleimola et al. (2010) | SMC 2010; conference | DOI 10.5281/zenodo.849811 | Zenodo and Maynooth repository records | Conference; not counted |
| Li et al. (2023) | Cognitive Computation; journal | DOI 10.1007/s12559-022-10031-5 | University repository records (vol. 15, pp. 23–35) | Not verified — manual database verification required. |
| Liu et al. (2025) | Preprint | arXiv:2509.24286 | arXiv listing (versions v1 and v2); venue not confirmed | Preprint; not counted |
| Masuda & Saito (2023) | IEEE/ACM TASLP; journal | DOI not retrieved | dblp and a reference list (pages 863–875) | Not verified — manual database verification required. |

| Reference | Venue and type | Identifier | Checked against | Scopus / WoS status |
|---|---|---|---|---|
| Mauch & Dixon (2014) | ICASSP 2014; conference | Not retrieved | Title seen on a teaching page; pages from the team's draft (†) | Conference; not counted |
| Raffel et al. (2014) | ISMIR 2014; conference | Not retrieved | Team's draft only (†) | Conference; not counted |
| Rogozinsky et al. (2022) | Proc. Telecommunication Universities; journal | Not retrieved | Journal site abstract; volume and pages from the supplied inventory (†) | Not verified — manual database verification required. |
| Rottondi et al. (2016) | IEEE Access; journal | DOI 10.1109/ACCESS.2016.2628440 | DOAJ record and repository abstract | Not verified — manual database verification required. |
| Tanaka et al. (2023) | Computer Music J.; journal | DOI 10.1162/comj_a_00672 | University repository records | Not verified — manual database verification required. |
| Välimäki & Huovilainen (2007) | IEEE Signal Process. Mag.; journal | Not retrieved | Year from a citing paper; pages from the team's draft (†) | Not verified — manual database verification required. |
| Välimäki et al. (2010) | IEEE/ACM TASLP; journal | DOI 10.1109/TASL.2009.2026507 | CCRMA publication page (start page 786; end page †) | Not verified — manual database verification required. |
| Zhang et al. (2026) | TISMIR; journal | DOI 10.5334/tismir.267 | Publisher XML and author's university page (vol. 9, pp. 102–118) | Not verified — manual database verification required. |

## E.2 Corrections to the supplied inventory and the earlier draft

The checks above found several inconsistencies. Items from the supplied inventory that could not be located are excluded from the reference list until found. Performance figures quoted in the inventory (accuracy percentages, F-measures, opinion scores and latency values) have not been checked against the papers and are not used in this proposal.

**Table E2. Corrections and exclusions.**

| Item | Supplied or earlier value | Checked record | Action |
|---|---|---|---|
| Li et al. | Cognitive Computation 14(6), 2111–2126 | Cognitive Computation 15(1), 23–35 (published online June 2022) | Corrected |
| Masuda & Saito | Pages 852–864 | Pages 863–875 (dblp and a citing reference list) | Corrected |
| Brum & Moreno | Pages 65–72 | Pages 50–57 with DOI | Corrected |
| Dong et al. | Pages 952–956 | dblp lists pages 951–955 | Corrected, to confirm |
| Liu et al. | ICASSP 2025 | Found as arXiv preprint 2509.24286, first submitted September 2025 and retitled between versions | Cited as preprint; venue unverified |
| Stillings (2021) thesis | UPF master's thesis | No matching record found in searches | Excluded until located |
| Latake et al. (2024) | Multimedia Research 7(1) | No matching record found; incomplete author data | Excluded; Benetos et al. (2013) used for transcription limits |
| Mauch & Dixon (2014), team draft | Title given as "pYIN: A significance-based probabilistic approach to YIN…" | Title listed elsewhere as "pYIN: A fundamental frequency estimator using probabilistic threshold distributions" | Title corrected, to confirm |
| Välimäki & Huovilainen, team draft | Year 2006 | Cited as 2007 by a later paper; issue 24(2) | Year corrected, to confirm |
| Inventory narrative | Sub-20 ms end-to-end latency; four synthesis engines; research gap and contribution claims | Not supported by any measurement or search; conflicts with the project rules and the frame-length argument in Section 2.5 | Not used |

## E.3 Source-to-design mapping

**Table E3. Which sources inform which decisions.**

| Source | What it supports | Design decision informed |
|---|---|---|
| Kleimola et al. (2010) | Discontinuous oscillator waveforms alias; PolyBLEP modifies samples around each discontinuity | PolyBLEP sawtooth and square oscillators; aliasing test T-S2 |
| Välimäki & Huovilainen (2007); Välimäki et al. (2010) | Antialiasing oscillators for subtractive synthesis; differentiated polynomial waveforms | Background for oscillator choice; DPW as a possible comparison |
| Rogozinsky et al. (2022) | Formal mathematical models of classical synthesis methods | Notation and structure of Section 4.3 |
| De Poli (2023); Zhang et al. (2026) | Taxonomy of synthesis models; evaluation dimensions for interactive synthesis | Method classification (Section 4.2); evaluation framing (responsiveness, controllability) |
| Chowning (1973) | FM synthesis and sideband structure | Optional FM extension; test T-S6 |
| de Cheveigné & Kawahara (2002) | YIN structure, reported error rates, low latency, few parameters | Primary estimator; Eqs. (13)–(15) |
| Mauch & Dixon (2014) | Probabilistic extension of YIN | Optional offline comparison |
| Benetos et al. (2013) | Limits of transcription; tailoring to use cases | Monophonic scope; limitations in Section 5.4 |
| Raffel et al. (2014) | Common MIR metric definitions | Pitch and note metrics in Appendix C |
| Rottondi et al. (2016) | Latency tolerance for synchronous performance | Latency discussion; no acceptance thresholds taken from it |
| Tanaka et al. (2023) | MIDI interface as boundary between sensing and synthesis | Event-bus architecture (learning-based mapping not used) |

### E.4 Search plan for the Fall week 10 literature survey

Databases: Scopus, Web of Science, IEEE Xplore, ACM Digital Library and Google Scholar for discovery. The primary window is 2016–2026, with older papers only where needed as mathematical foundations (for example YIN). Inclusion: peer-reviewed journal articles in English, indexed in Scopus and/or Web of Science (checked through the databases themselves), directly relevant to the topics below. Exclusion: preprints, theses and conference papers for the 25-paper quota; learning-based methods except as a short, labelled context section. Targets are provisional and total 25.

**Table E4. Planned searches and provisional targets.**

| Topic | Example search string | Target | Note |
|---|---|---|---|
| Antialiased oscillators and virtual analog | ("virtual analog" OR "subtractive synthesis") AND (aliasing OR bandlimited OR PolyBLEP OR "differentiated polynomial") | 5 | Also filter design and digital filters |
| Classical f0 estimation | ("fundamental frequency" OR pitch) AND (estimation OR tracking) AND (autocorrelation OR YIN OR AMDF OR "difference function") | 7 | Include noise robustness and octave errors |
| Whistling and humming pitch | (whistle OR whistling OR humming) AND pitch AND (detection OR estimation OR "query by humming") | 3 | Few papers expected; relax if needed |
| Onset detection and note segmentation | ("onset detection" OR "note segmentation") AND (music OR audio) | 4 | Classical methods preferred |
| Audio-to-MIDI and evaluation | ("audio-to-MIDI" OR "music transcription") AND (evaluation OR metrics) | 3 | Use as scope context |
| Real-time audio and latency | ("audio latency" OR "real-time audio") AND (software OR synthesizer OR "audio interface") | 3 | Measurement methods |

## Appendix F. Open Decisions

**Table F1. Decisions awaiting confirmation.**

| ID | Decision | Current assumption | To be confirmed by |
|---|---|---|---|
| D1 | Primary synthesis method | Subtractive (PolyBLEP, biquad, ADSR); FM and wavetable are staged optional extensions | Supervisor |
| D2 | Project title | MidiKlavye: a digital subtractive software synthesizer with a classical-DSP pitch-to-MIDI interface | Supervisor |
| D3 | Real-time microphone input as core or stretch | Should (stretch), with offline fallback | Team and supervisor |

| ID | Decision | Current assumption | To be confirmed by |
|---|---|---|---|
| D4 | Optional extensions (FM, wavetable, hardware keyboard, GUI, MIDI Learn) | Attempted only if the primary chain passes the TW7 checkpoint | Team and supervisor |
| D5 | Faculty name on the cover page | Wording of the team's earlier draft; the guidelines print a different word order | Supervisor, or the official template |
| D6 | Order of appendices and references | Appendices before References, as in the guidelines' proposal format; the guidelines also contain the opposite statement | Supervisor |
| D7 | Journal quota: which sources count and how borderline cases are treated | Only database-verified journal articles count | Supervisor |
| D8 | Recordings of persons other than the team | Not planned; would need guidance on approval | Supervisor |
| D9 | Mapping of week numbers to calendar dates | Fall ends in January; Spring ends on 30 April 2027 | Academic calendar |
| D10 | Package leads for each team member | Joint work; leads to be recorded in the interim report | Team |
| D11 | Sampling rate and block size | Chosen after the device test in WP1 | Team |
| D12 | Weekly roadmap versus the official 16- and 14-week calendar | Fall TW1–TW12 in official weeks 5–16; Spring SW1–SW12 in weeks 1–12 with weeks 13–14 for corrections and presentation | Supervisor |
| D13 | CREPE in the roadmap (Fall TW1 reading, Spring SW3 integration) | Read as context only; not integrated, because of the classical-only restriction | Supervisor |
| D14 | Order of subtractive synthesis, polyphony and the optional FM work (TW5, TW6, TW8) | Subtractive synthesis and polyphony first, because they are core; FM follows the TW7 checkpoint | Team and supervisor |
| D15 | Results deadline (Spring week 8) before systematic experiments (SW9) | Week 8 report: design, implementation and preliminary results; full experiments in the discussion | Supervisor |

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*† Some bibliographic details marked † come from secondary listings or the team's earlier documents and must be confirmed against the publisher page before submission (see Appendix E).*